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* Sp. that we have a testing problem

$$(H) = (H_0) \cup (H_A) \quad , \quad (H_0) \cap (H_A) = \emptyset$$

$$H_0: \theta \in (H_0) \quad \text{v/s} \quad H_A: \theta \in (H_A)$$

Observation : $X \sim f(x|\theta) \leftarrow \text{pdf/pmf}$

Let T be a suff. stat. for θ . Then it is enough to consider only test func's that are func's of T

Recall $\phi: \mathcal{X} \rightarrow [0,1]$ is a test func'.

$\phi(x)$ = prob. of rejecting H_0 if we observe $X=x$

Let $\psi(T) = E(\phi|T)$ = "a test func" that is a func' of T and because $E_\theta(\psi) = E_\theta(\phi)$, ψ & ϕ have same prob. of $T1$ & $T2$ errors.

→ Special case: $(H_0) = \{\theta_0\}$, $(H_A) = \{\theta_1\}$

$$H_0: \theta = \theta_0 \quad \text{v/s} \quad H_A: \theta = \theta_1$$

$$T(x) = \frac{f(x|\theta_1)}{f(x|\theta_0)} \quad (\text{Assume that the support of } X \text{ doesn't depend on } \theta)$$

• We may reject H_0 if $\frac{f(x|\theta_1)}{f(x|\theta_0)} > 1$

& accept H_0 if $\frac{f(x|\theta_1)}{f(x|\theta_0)} < 1$

$$\left(\text{Assume } P_\theta \left\{ \frac{f(x|\theta_1)}{f(x|\theta_0)} = 1 \right\} = 0 \quad \forall \theta \right)$$

- This test has max. power among all tests having the same size. (It needs a ~~test~~ proof)

The proof is exercise (dr.)

Let $\psi(x)$ be indicator for whether $LR > 1$ or not. Then look at:
 $(\psi(x) - \phi(x))(f_1(x) - f_0(x))$. Find an inequality and integrate.

$X \sim \text{Bin}(n, \theta)$, $0 < \theta < 1$

Beta family of priors:

$$\pi(\theta) = \frac{\theta^{m-1} (1-\theta)^{n-1}}{\beta(m, n)}, \quad m, n > 0, \quad 0 < \theta < 1.$$

Calculate the posterior: $\pi(\theta|x)$

$$f(x|\theta) = \binom{n}{x} \theta^x (1-\theta)^{n-x}$$

$$\left(\text{Apparently: } \pi(\theta|x) = \frac{\theta^{m+x-1} (1-\theta)^{n+r-x-1}}{\beta(m+x, n+r-x)} \right)$$

$X \sim \text{Poi}(\theta)$, $\theta > 0$

$$f(x|\theta) = \frac{e^{-\theta} \theta^x}{x!}$$

Can you guess a conjugate family of priors distribution $\pi(\theta)$?

$$\left(\text{Apparently, } \pi(\theta) = \frac{\theta^{\alpha-1} \cdot e^{-\theta/\lambda}}{\Gamma(\alpha) \cdot \lambda^\alpha} \right)$$

$X \sim N(\theta, \sigma^2) \rightarrow \sigma^2 \text{ known}, \theta \in \mathbb{R}$
 $\hookrightarrow \text{unknown}$

What is a conjugate family of priors $\pi(\theta)$?

(Apparently it'll be $N(\mu, \tau^2)$, τ is anything)

Identify a conjugate family for the Cauchy distⁿ.

→ Bayes estimate is expectation of posterior distⁿ. $\neq E(\theta|X)$

Q) Is the Bayes estimate unb. for θ in these cases?

(1) Binomial & Beta prior (Apparently no)

(2) Poisson & Gamma prior (Apparently no)

(3) ~~Normal~~ Normal & Normal prior (Apparently no)

Q) $X \sim f(X|\theta)$, $\pi(\theta)$ is a prior (not necessarily conjugate)
Can Bayes estimate $\hat{\theta}^T$ be an unbiased estimate for θ ?

• some calculation:

$$m(X|\theta) = f(X|\theta) \cdot \pi(\theta)$$

$$E_m \{T - \theta\}^2 = E_m \{T^2 + \theta^2 - 2T\theta\}$$

Observe: $E(T\theta) = E(\theta^2) = E(T^2)$ IF T is unb.

$$= E(E(T\theta|\theta)) = E(E(T\theta|X))$$

$$= E[\theta E(T|\theta)] = E[T E(\theta|X)] \Rightarrow E_m[(T - \theta)^2] = 0$$

$$= E[\theta^2] = E[T^2]$$

A) NO, ~~ex~~ except in trivial degenerate cases where $E_m[(T - \theta)^2] = 0$.

→ $f(X|\theta) = g(\theta, T(x)) \cdot h$, $\pi(\theta)$ is prior.

$$\pi(\theta|X) = \frac{g(\theta, T(x)) \cdot h \cdot \pi(\theta)}{\int g(\theta, T(x)) \cdot h \cdot \pi(\theta) d\theta} = \frac{g(\theta, T(x)) \cdot \pi(\theta)}{\int g(\theta, T(x)) \cdot \pi(\theta) d\theta}$$

↳ The posterior is a funcⁿ of the suff. stat., if there is a prior and a suff. stat.